

Intensity Measure

ImageJ ROI Intensity Analysis Script

Documentation & User Guide

Version 2.1

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Abstract

A JavaScript macro for ImageJ/Fiji that measures fluorescence intensity across image stacks with **multiple normalization methods**. Suitable for time-lapse imaging, photobleaching studies, calcium imaging, FRAP, and general fluorescence quantification. This script offers six different normalization methods including F/F_0 , $\Delta F/F_0$, Z-score, and more.

Contents

1 Overview

This script automates the process of measuring fluorescence intensity changes across a time-lapse image stack. It offers **six different normalization methods** to suit various experimental needs, from photobleaching analysis to calcium imaging to protein expression quantification.

1.1 Use Cases

- **Photobleaching / FRAP:** Track fluorescence decay or recovery over time
- **Calcium imaging:** Monitor intracellular calcium dynamics
- **Drug response:** Measure intensity changes upon treatment
- **Expression analysis:** Compare fluorescence levels across conditions
- **Time-lapse studies:** Quantify any temporal intensity changes

2 Features

- **Six normalization methods:** F/F_0 , $\Delta F/F_0$, background subtraction, percent of max, Z-score, or raw values
- **Flexible background selection:** Click on the image, type coordinates, or skip background correction
- **Multiple ROI tools:** Rectangle, Oval, Freehand, Polygon, Brush, or use existing ROI
- **Per-frame background correction:** Accounts for fluctuating background
- **CTCF calculation:** Corrected Total Cell Fluorescence for every frame
- **Validation checks:** Ensures selections are within image bounds
- **Results table output:** Easy export to CSV for further analysis

3 Installation

3.1 Requirements

- [ImageJ](#) (v1.53+) or [Fiji](#)

3.2 Setup

1. Download `Intensity_measure.js`
2. Place it in your ImageJ plugins folder:
 - **Windows:** `C:\Program Files\ImageJ\plugins\`
 - **macOS:** `/Applications/ImageJ.app/plugins/`
 - **Linux:** `~/ImageJ/plugins/`
3. Restart ImageJ, or use `Plugins > Macros > Run...` to execute directly

4 Usage

4.1 Step-by-Step Guide

4.1.1 Step 1: Open Your Image Stack

Open your time-lapse fluorescence image stack in ImageJ:

- File > Open... or drag-and-drop
- Ensure it's a stack (multiple frames), not a single image

4.1.2 Step 2: Run the Script

- Plugins > Macros > Run... → Select `Intensity_measure.js`
- Or if installed: Plugins > `Intensity_measure`

4.1.3 Step 3: Select Normalization Method

Choose from six normalization options:

Method	Output Range	Best For
F/F ₀ (default)	Starts at 1.0	Photobleaching, FRAP, general time-lapse
$\Delta F/F_0$	Starts at 0.0	Calcium imaging, stimulus response
Background subtraction	Absolute units	Simple correction, comparing intensities
Percent of maximum	0–100%	Normalizing across experiments
Z-score	Mean=0, SD=1	Statistical comparison, outlier detection
Raw values	Original units	No processing needed

Table 1: Available normalization methods

4.1.4 Step 4: Select Background Point

You'll see a dialog asking how to select the background reference:

Option	Description
Click to select	Use the Point tool to click on a background region
Type coordinates	Enter exact X, Y pixel coordinates
No background correction	Skip background subtraction entirely

Tips for background selection:

- Choose a region with NO fluorescent signal
- Avoid dark artifacts or dead pixels
- The same pixel location is used across all frames

4.1.5 Step 5: Select Target ROI

Choose your preferred ROI selection method:

Option	Description
Rectangle	Draw a rectangular selection
Oval	Draw an elliptical selection
Freehand	Draw a freeform closed shape
Polygon	Click vertices to create a polygon
Brush	Paint the selection with a brush
Use existing ROI	Use a previously drawn ROI on the image
Type coordinates (Rectangle)	Enter X, Y, Width, Height for a rectangle
Type coordinates (Oval)	Enter X, Y, Width, Height for an oval

4.1.6 Step 6: View Results

A Results Table will appear with measurements for each slice:

Slice	Raw_Mean	Background	F_corrected	F/F0	CTCF
1	1500.0	100.0	1400.0	1.0000	56000.0
2	1450.0	102.0	1348.0	0.9629	53920.0
3	1380.0	99.0	1281.0	0.9150	51240.0

4.1.7 Step 7: Export Data

- File > Save As... in the Results window
- Save as .csv for Excel, Python, R, MATLAB, etc.

5 Normalization Methods

5.1 F/F₀ (Normalize to First Frame)

Formula:

$$\frac{F}{F_0} = \frac{F(t)}{F(t_0)} \quad (1)$$

- Output starts at **1.0** and changes from there
- Standard for **photobleaching**, **FRAP**, general time-lapse
- Easy to interpret: 0.5 = 50% of initial intensity

5.2 ΔF/F₀ (Fold Change from Baseline)

Formula:

$$\frac{\Delta F}{F_0} = \frac{F(t) - F(t_0)}{F(t_0)} \quad (2)$$

- Output starts at **0.0** (baseline)
- Standard for **calcium imaging**, stimulus-response experiments
- Positive values = increase, negative = decrease
- A value of 1.0 means 100% increase (doubled)

5.3 Background Subtraction Only

Formula:

$$F_{\text{corrected}} = F(t) - F_{\text{background}}(t) \quad (3)$$

- Returns absolute intensity values (background-corrected)
- Good for comparing intensities within the same experiment
- No normalization to baseline

5.4 Percent of Maximum

Formula:

$$\% = \frac{F - F_{\min}}{F_{\max} - F_{\min}} \times 100 \quad (4)$$

- Scales data to **0–100%** range
- Useful for comparing across different experiments
- Min intensity = 0%, Max intensity = 100%

5.5 Z-score Normalization

Formula:

$$Z = \frac{F - \mu}{\sigma} \quad (5)$$

- Mean becomes 0, standard deviation becomes 1
- Good for **statistical comparisons** across experiments
- Identifies outliers ($|Z| > 2$ or 3)

5.6 Raw Values (No Normalization)

- Returns the original mean intensity values
- No background correction or normalization applied
- Use when you need unprocessed data

6 Output Columns Explained

7 Mathematical Background

7.1 F/F₀ Normalization

The **F/F₀** ratio normalizes all measurements to the first frame:

$$\frac{F}{F_0} = \frac{F(t) - F_{bg}(t)}{F(t_0) - F_{bg}(t_0)} \quad (6)$$

Where:

- $F(t)$ = Mean fluorescence intensity at time t
- $F_{bg}(t)$ = Background intensity at time t
- $F(t_0)$ = Mean fluorescence at the first frame
- $F_{bg}(t_0)$ = Background at the first frame

Column	Formula	Description
Slice	—	Frame number (1-indexed)
Raw_Mean	$\text{mean}(\text{ROI pixels})$	Uncorrected mean intensity within ROI
Background	$\text{pixel}(x_{ref}, y_{ref})$	Background value at reference point (if used)
F_corrected	$\text{Raw_Mean} - \text{Background}$	Background-subtracted mean intensity
	Varies by method	Normalized > Normalized The normalized value (column name depends on method)
CTCF	$(A \times \bar{I}) - (A \times \bar{I}_{bg})$	Corrected Total Cell Fluorescence

7.2 $\Delta F/F_0$ (Fold Change)

$$\frac{\Delta F}{F_0} = \frac{F(t) - F_0}{F_0} = \frac{F}{F_0} - 1 \quad (7)$$

This is simply F/F_0 minus 1, shifting the baseline to 0.

7.3 CTCF (Corrected Total Cell Fluorescence)

CTCF accounts for both intensity and area:

$$\text{CTCF} = (A_{\text{ROI}} \times \bar{I}_{\text{ROI}}) - (A_{\text{ROI}} \times \bar{I}_{\text{background}}) \quad (8)$$

This is useful when comparing absolute fluorescence amounts across different sized structures.

7.4 Exponential Decay Fitting

For photobleaching analysis, fit your F/F_0 data to:

$$\frac{F}{F_0} = A \cdot e^{-t/\tau} + B \quad (9)$$

Where:

- τ = photobleaching time constant
- A = bleachable fraction
- B = non-bleachable baseline

Half-life calculation:

$$t_{1/2} = \tau \cdot \ln(2) \approx 0.693 \cdot \tau \quad (10)$$

8 Best Practices

8.1 For Time-Lapse Experiments

1. Consistent imaging conditions

- Same exposure time for all frames
- Same excitation intensity
- Let laser/lamp stabilize before imaging

2. Background selection

- Choose a region that never contains signal
- Avoid edges of the field of view
- Check that background doesn't change dramatically

3. ROI selection

- Include the entire fluorescent structure
- Don't include neighboring cells/structures
- Use the same ROI for all comparisons

8.2 Choosing a Normalization Method

Experiment Type	Recommended Method
Photobleaching	F/F_0
FRAP	F/F_0
Calcium imaging	$\Delta F/F_0$
Drug response	$\Delta F/F_0$ or F/F_0
Comparing expression levels	Background subtraction or CTCF
Cross-experiment comparison	Percent of max or Z-score
Raw data export	Raw values

Table 2: Recommended normalization methods by experiment type

9 Troubleshooting

9.1 “Initial ROI intensity is $\leq$ background”

Cause: Your selected ROI has lower intensity than the background reference point.

Solution:

- Check that the background point is truly in a dark region
- Ensure the ROI contains fluorescent signal
- Try “No background correction” option
- Verify you're on the correct image/channel

9.2 “No ROI selected”

Cause: You clicked OK without drawing a selection.

Solution:

- Draw the ROI first, then click OK
- For “Use existing ROI”: ensure an ROI exists before running the script

9.3 “Coordinates out of image bounds”

Cause: Typed coordinates exceed image dimensions.

Solution:

- Check image size: **Image > Properties**
- Coordinates are 0-indexed (0 to width-1, 0 to height-1)

9.4 Results show NaN or Infinity

Cause: Division by zero ($F_0 = 0$) or invalid measurements.

Solution:

- Background point might be inside a bright region
- Try “Background subtraction only” method
- Check for corrupted frames in the stack

9.5 Unexpected normalized values

Cause: Intensity changes not matching expectations.

Solution:

- Verify background point is in a non-fluorescent region
- Check for photobleaching during setup
- Ensure first frame is representative baseline

10 Example Workflow

1. Open stack: **File > Open > my_timelapse.tif**
2. Run script: **Plugins > Macros > Run > Intensity_measure.js**
3. Normalization: Select “F/F₀ (Normalize to first frame)”
4. Background: Click to select → Click on dark region → OK
5. ROI: Freehand → Trace around cell → OK
6. Results table appears
7. Export: **File > Save As > intensity_data.csv**
8. Analyze/plot in your preferred software

11 File Structure

```
Ares/  
  Align_Stack.js  
  Crop_Stack_Subsets.js  
  Intensity_measure.js  
  Intensity_measure_documentation.tex  
  README.md  
  CHANGELOG.md
```

```
CITATION.cff
docs/
  intensity-measure.md
  align-stack.md
  stack-crop-subsets.md
```

12 Changelog

- v1.0** Initial release with basic rectangle ROI and point background
- v1.1** Added coordinate input options
- v1.2** Added F/F_0 photobleaching normalization
- v1.3** Added multiple ROI selection tools (oval, freehand, polygon, brush)
- v2.0** Added multiple normalization methods (F/F_0 , $\Delta F/F_0$, background subtraction, percent max, Z-score, raw)
- v2.1** Corrected CTCF area handling for non-rectangular ROIs and updated repository documentation structure

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